Problem 5 - Package Drop-Off Log

Week 7 | Category B Practice | Java solutions

Problem

DeliveryNote declares abstract confirmDelivery() and overloads it with a signature parameter. ParcelNote and LetterNote directly extend it.

Complete Java Code

```
abstract class DeliveryNote {
    public abstract String confirmDelivery();

    public String confirmDelivery(String signature) {
        return confirmDelivery() + ", signed by " + signature;
    }
}

class ParcelNote extends DeliveryNote {
    private final String trackingId;
    public ParcelNote(String trackingId) { this.trackingId = trackingId; }
    @Override public String confirmDelivery() {
        return "Parcel " + trackingId + " delivered";
    }
}

class LetterNote extends DeliveryNote {
    private final String trackingId;
    public LetterNote(String trackingId) { this.trackingId = trackingId; }
    @Override public String confirmDelivery() {
        return "Letter " + trackingId + " delivered";
    }
}

public class Main {
    static void logAll(DeliveryNote[] notes) {
        for (DeliveryNote note : notes) {
            System.out.println(note.confirmDelivery());
        }
    }

    public static void main(String[] args) {
        ParcelNote p = new ParcelNote("TRK-1");
        System.out.println(p.confirmDelivery());
        System.out.println(p.confirmDelivery("J. Smith"));
        DeliveryNote ref = p;
        logAll(new DeliveryNote[] { ref, new LetterNote("TRK-2") });
    }
}
```

Sample Output

```
Parcel TRK-1 delivered
Parcel TRK-1 delivered, signed by J. Smith
Parcel TRK-1 delivered
Letter TRK-2 delivered
```

How it works

The overloaded method adds signature details by calling confirmDelivery() once, keeping the base message in the subtype. logAll() demonstrates runtime polymorphism through DeliveryNote references.

Source basis: Week 7 Category B assignment/practice PDF. Examples and requirements are preserved; implementation details are filled in to create a complete runnable Java example.